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Liver Metabolism and Production of Cows Fed Increasing Amounts of Rumen-Protected Choline During the Periparturient Period[1]

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ABSTRACT

Forty-eight multiparous Holstein cows were fed treatments consisting of either 0, 45, 60, or 75 g/d of a rumen-protected choline (RPC) source in a completely randomized design from 21 d before expected calving to 63 d postpartum to determine whether choline supplementation to the diet would affect hepatic fatty acid and glucose metabolism, key metabolites in plasma, and cow performance. Dry matter intake (DMI), milk yield, body condition score, and body weights (BW) were similar for cows receiving the four treatments. Feeding RPC tended to increase yields of milk fat, 3.5% fat-corrected milk, and total solids. Plasma concentrations of nonesterified fatty acids and β-hydroxybutyrate were not different among cows fed the four treatments. Concentrations of triglycerides in liver were similar, but concentrations of glycogen in liver increased as cows consumed increasing amounts of RPC. Hepatic capacity for storage of [1-^{14}C]palmitate as esterified products within liver slices tended to decrease as the amount of RPC consumed by cows increased; however, effects of treatment on hepatic capacity for oxidation of [1-^{14}C]palmitate to CO_2 were not significant. These data imply that choline may increase the rate of very low density lipoprotein synthesis and secretion of esterified lipid products from liver. Hepatic capacities for conversion of [1-^{14}C] propionate to CO_2 and to glucose in liver were similar among cows fed the four treatments. Collectively, these results suggest that hepatic fatty acid metabolism and cow performance are responsive to increasing the supply of choline during the periparturient period.
(**Key words:** dairy cow, choline, gluconeogenesis, hepatic lipidosis)

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Abbreviation key: MUN = milk urea N; **RPC** = rumen-protected choline; **VLDL** = very low density lipoproteins.

INTRODUCTION

The periparturient period of dairy cattle is characterized by dramatic changes in nutrient demand that necessitate remarkable coordination of metabolism to meet requirements for energy, glucose, and AA by the mammary gland following calving (Bell, 1995). To meet their energy requirements during the periparturient period and early lactation, dairy cows mobilize large amounts of fatty acids from adipose tissue, resulting in increased circulating concentrations of NEFA in the bloodstream. Although NEFA can be used by other tissues for energy and for milk fat, the liver typically extracts NEFA in proportion to its supply (Emery, 1992). Hepatic capacities for fatty acid oxidation and export as very low density lipoproteins (**VLDL**) are low in ruminants (Grummer, 1993); therefore, excessive uptake of NEFA by the liver can lead to the development of hepatic lipidosis (fatty liver caused by the accumulation of triglycerides within the liver parenchyma). Furthermore, it has been demonstrated that triglyceride accumulation in the liver reduces the capacity of the liver to detoxify ammonia to urea (Strang et al., 1998), which, in turn, may affect gluconeogenic capacity from propionate, the predominant glucogenic precursor (Overton et al., 1999).

Choline is considered a quasi-vitamin due to its necessity for maintaining health in certain species or when methyl precursors are low in the diet (Combs, 1992). As reviewed by Sharma and Erdman (1989), choline must be supplemented to chicks, laying hens, gestating sows, and preruminant lambs and calves. Choline deficiency in rats has been shown to cause more than a sixfold increase in the accumulation of triglycerides in the liver (Yao and Vance, 1990), which perhaps is similar to what occurs in the periparturient dairy cow. It is our hypothesis that choline may be beneficial for the periparturient dairy cow because of its roles in methyl donation and phospholipid formation for lipoprotein metabolism. Like other species, the dairy cow is capable of synthesizing phosphatidylcholine either from the tri-methylation of phosphatidylethanolamine originally derived from serine, or activation of choline to cytadine diphosphate-choline and addition of a 1,2-diglyceride (Combs, 1992). It is conceivable that the decreased DMI occurring before parturition (Grummer, 1995) may result in decreased supply of dietary precursors of phosphatidylcholine, leading to increased risk for development of hepatic lipidosis during the immediate peripartal period.

If synthesis of choline and related compounds during the periparturient period is insufficient for maximal hepatic metabolism of NEFA, the severity of hepatic lipid infiltration may be exacerbated by providing additional choline through the diet. The benefit could be similar to the effects reported when hepatocytes isolated from rats fed a choline-free diet were supplemented with choline or Met in vitro (Yao and Vance, 1988). Supplementation with either nutrient increased concentrations of phosphatidylcholine in media and export of triglycerides from the hepatocytes as VLDL.

The microbial populations in the rumen quickly degrade dietary choline; therefore, the only practical means of increasing choline to the periparturient dairy cow is to feed it in a rumen-protected form (Atkins et al., 1988). Pinotti et al. (2000) fed 20 g/d of a rumen-protected choline (**RPC**) source (~5 g/d of choline chloride) to dairy cows starting 14 d before expected calving date through 30 DIM. They used an index for development of fatty liver (ratio of plasma concentrations of NEFA to cholesterol) and determined that this index was reduced, and cows produced more milk when fed diets supplemented with choline. Hartwell et al. (2000) fed RPC (0, 24, and 48 g/d to provide 0, 6, and 12 g/d choline chloride) to periparturient dairy cattle in conjunction with either a diet containing 4.0% RUP or 6.2% RUP and reported no significant differences in liver triglyceride content attributable to choline supplementation. There was, however, an increase in milk yield when choline was supplemented to cows fed the low RUP diet, indicating that perhaps the amount of AA supplied through the diet can influence the effectiveness of RPC supplementation.

Our hypothesis is that if choline supplementation decreases the accumulation of triglycerides in liver, the capacity of liver to synthesize glucose should increase. The objective of this experiment was to determine the effects of feeding increasing amounts of RPC to periparturient cows on hepatic fatty acid and glucose metabolism, key metabolites in plasma, and cow performance.

MATERIALS AND METHODS

The Cornell University Institutional Animal Care and Use Committee approved all procedures involving animals. Forty-eight Holstein dairy cows entering second or greater lactation were moved into individual tie stalls ~28 d before expected parturition and fed for ad libitum intake the same prepartum diet (Table 1) once daily as TMR. On d 21 before expected parturition, cows were assigned to one of four amounts (0, 45, 60, or 75 g/d) of RPC (Reashure, Balchem Encapsulates, Slate Hill, NY). The RPC contained 25% choline chloride (wt/wt) which is approximately 85% rumen-protected (Deuchler et al., 1998). Treatment assignments were essentially random. Because prepartum BCS has been shown to have a relationship with the amount of accumulation of triglyceride in liver postpartum (Hartwell et al., 2000) and level of milk production may affect production parameters and energy balance, pretreatment BCS and PTA milk were used as criterion for treatment assignments for approximately the final 12 cows assigned to treatments. The RPC was top dressed onto the TMR immediately postfeeding and mixed manually into the top 10 cm of feed. After parturition, cows were fed a postpartum diet (Table 1), and supplementation with RPC was continued for appropriate cows until 63 d of lactation. Because Met can contribute to choline synthesis (Emmanuel and Kennelly, 1984), both prepartum and postpartum diets contained corn gluten meal to increase Met supply, thereby reducing the impact that feeding a diet low in Met supply could have had on the outcome of this study. Amounts of feed offered and refused were measured daily throughout the experiment, and weekly analyses of DM content of the TMR were used to calculate DMI. Samples of feed and TMR were collected on a weekly basis and dried to static weight at 60°C. Samples were ground through a 2-mm screen in a Wiley Mill, and monthly composites were prepared. Monthly composites of forages and total

Table 1. Ingredient and chemical composition of pre- and postpartum diets (% of DM).

Item	Prepartum	Postpartum
Corn silage	24.1	27.0
Hay-crop silage	29.9	27.0
Grass hay	8.9	—
High moisture corn	16.7	22.0
Expeller soybean meal	6.7	4.4
Soybean meal	3.3	4.9
Soy hulls	5.4	4.7
Corn gluten meal	2.7	4.0
Cottonseeds, whole	—	2.9
Calcium carbonate	0.44	—
Magnesium sulfate	0.53	—
Salt	0.22	—
Sodium bicarbonate	—	0.76
Mineral and vitamin mix[1]	1.01	2.45

[1]The mineral and vitamin mix for prepartum cows contained approximately 18% Ca, 6% P, 9% Mg, 0.8% K, 39% Na, 3% S, 5000 ppm Fe, 5000 ppm Zn, 1800 ppm Cu, 5000 ppm Mn, 148 KIU/kg of vitamin A, 37 KIU/kg of vitamin D, and 1481 KIU/kg of vitamin E. The mineral and vitamin mix for postpartum cows contained approximately 20% Ca, 5% P, 3% Mg, 1% K, 7.75% Na, 2% S, 3500 ppm Fe, 2250 ppm Zn, 500 ppm Cu, 2250 ppm Mn, 55.5 KIU/kg of vitamin A, 13.9 KIU/kg of vitamin D, and 347.2 KIU/kg of vitamin E.

Table 2. Chemical composition of pre- and postpartum diets (% of DM).

Item	Prepartum	Postpartum
DM	44.9	44.3
CP	14.80	17.84
Acid detergent insoluble CP	0.94	1.05
Neutral detergent insoluble CP	3.29	3.00
Soluble CP	38.5	41.8
ADF	26.9	22.0
NDF	43.8	34.7
Lignin	4.85	4.43
Non-structural carbohydrate	30.8	35.3
Nonfiber carbohydrate[1]	33.7	37.3
Crude fat	4.66	5.05
Ash	6.35	8.15
NE_L, Mcal/kg	1.51	1.64
Ca	0.64	0.93
P	0.36	0.52
Mg	0.36	0.30
K	1.26	1.22
Na	0.28	0.42
S	0.24	0.24
Fe, ppm	139.2	215.4
Cu, ppm	14.3	18.2
Zn, ppm	48.0	81.0
Mn, ppm	49.10	75.94
Mo, ppm	1.22	1.30

[1]Calculated according to National Research Council (2001).

experiment composites of concentrate ingredients were analyzed (Dairy One Laboratories, Ithaca, NY) using wet chemistry techniques for DM, OM, CP, soluble CP, ADF, NDF, EE, ADFCP, NDFCP, lignin, and minerals. Nutrient analyses for each ingredient are presented in Table 3.

Cows were milked three times per day at ~8-h intervals, and individual milk weights were measured at each milking. Milk was sampled from each milking during one 24-h period each week, composited, and analyzed for content of fat, CP, lactose, TS, and milk urea N (**MUN**) by midinfrared spectroscopy (Dairy One Laboratories, Ithaca, NY). A portion of milk composites from wk 2, 5, and 8 were frozen at −20°C until subsequent analysis for fatty acids.

Body weights of cows were measured on 1 d/wk after the morning milking and before feeding. Two individuals assigned BCS to cows on a weekly basis (1 = thin, 5 = fat; Wildman et al., 1982).

Blood samples were collected by venipuncture of the coccygeal vein/artery using heparinized Vacutainer tubes (Becton Dickinson, Franklin Lanes, NJ). Samples were obtained on 1 d during the 7 d before assignment to treatment, and then on Monday, Wednesday, and Friday of each week from approximately 21 d before parturition to 21 d postpartum and on d 28 and 35 after parturition. Sampling time (approximately 0900 h) corresponded to the time after orts were removed, after lactating cows returned from the parlor, and before feeding. Blood samples were placed on ice immediately fol-

Table 3. Results of wet chemistry analysis of dietary ingredients (DM basis).

Item	Corn silage	Grass silage (prepartum diet)	Grass hay	Mixed, mostly legume silage (postpartum diet)	High-moisture corn	Expeller soybean meal	Soybean meal	Soybean, Hulls	Corn gluten meal	Cottonseed, whole
CP, %	6.6	12.1	7.3	19.5	8.6	47.7	54.3	11.6	68.4	27.9
Acid detergent insoluble CP, %	0.73	1.07	1.00	1.54	0.4	1.4	1.3	1.2	3.1	1.6
Soluble protein, % of CP	54	47	22	58	41	4	17	21	5	23
Neutral detergent insoluble CP, %	1.3	3.5	3.0	4.3	0.9	16.8	2.7	4.4	3.8	2.4
ADF, %	26.0	37.5	45.0	37.8	3.3	10.9	8.2	48.6	5.9	32.2
NDF, %	45.3	58.5	70.5	53.4	8.5	27.2	10.3	66.5	10.1	44.5
Lignin, %	3.7	7.9	7.8	8.59	1.2	1.6	2.8	3	3.6	14.2
Nonfiber carbohydrate, %[1]	42.2	19.7	18.6	15.8	77.1	28.6	28.8	19.7	13.4	3.4
Crude fat, %	4.3	6.0	1.8	6.2	4.6	7	2	1.9	7.9	22.2
Ash, %	2.9	7.2	4.8	9.4	2.1	6.3	7.3	4.7	4.0	4.4
NE_L, Mcal/kg	1.63	1.26	0.90	1.28	2.09	2.07	1.83	1.46	1.90	2.16

[1]Calculated according to National Research Council (2001).

lowing collection. Plasma was harvested after centrifugation of the blood at 2060 × g for 15 min at 5°C. Plasma was stored at –20°C until subsequent analysis for NEFA (NEFA-C, WAKO, Dallas, TX as modified by McCutcheon and Bauman, 1986) and β-hydroxybutyrate (Kit 310-UV, Sigma Chemical Co., St. Louis, MO).

Liver samples were obtained by percutaneous trochar biopsy (Veenhuizen et al., 1991) from each cow on 1 d during the week before assignment to treatment, 1 d after calving, and again 21 d after calving. Total liver removed was ~3 g for each biopsy. Liver was blotted to remove excess blood and connective tissue. A portion of the liver was frozen in liquid N_2 for subsequent analysis for content of triglyceride and glycogen (see below). The remaining liver was transported to the laboratory in ice-cold PBS (0.02 m*M*; pH 7.4) and used immediately (within 75 min of biopsy) to determine hepatic capacities for fatty acid metabolism and gluconeogenesis using an in vitro metabolic incubation system (see below).

Milk Fatty Acid Composition

Milk was thawed followed by heating in a water bath to 37°C. Samples then were shaken and ~20 ml of milk was poured into plastic centrifuge tubes. Tubes then were placed into a cold centrifuge (5°C) and centrifuged for 30 min at 1790 × g. This caused the cream layer to float to the top. To clean the cream layer, 300 to 400 mg were placed into a glass extraction tube (16 × 150 mm, prerinsed with hexane), and extraction was carried out as described by Hera and Radin (1978). Approximately 30 mg of extracted lipid were weighed into tubes. Methyl esters of the fatty acids were prepared by transesterification with sodium methoxide according to the method of Christie (1982), as detailed by Chouinard et al. (1999).

Fatty acid methyl esters were quantified by GLC using a SP-2560 capillary column (100 m × 0.25 mm i.d. × 0.2-μm film thickness; Supelco, Inc., Bellefonte, PA). The analysis involved a programmed run with temperature ramps. The oven temperature was initially 50°C for 1 min then ramped to 160°C at 5°C/min and maintained for 42 min. The temperature was then ramped again at 5°C/min to 190°C and maintained for 22 min. Injector and detector temperatures were maintained at 250°C. The flow rate for hydrogen carrier gas was 1 ml/min. Hydrogen flow to the detector was 25 ml/min, airflow was 400 ml/min, and the nitrogen make-up gas flow was 45 ml/min.

Each peak was identified and quantified using pure methyl ester samples (Nu Chek Prep, Elysian, MN). A butter reference standard (CRM 164; Commission of the European Communities, Community Bureau of Reference, Brussels, Belgium) was used to determine recoveries and correction factors for individual fatty acids. The butter reference standard was also used at regular intervals throughout the analysis as an aid in quality control.

Liver Composition

Between 35 and 50 mg of liver tissue was used to determine content of glycogen according to the procedure of Lo et al. (1970). Liver triglyceride content was determined by first extracting ~400 mg of liver tissue according to the procedure of Hara and Radin (1978). The resulting hexane layer was removed to preweighed scintillation vials, and the hexane was removed by evaporation under a stream of N_2 gas at 37°C. Evaporated samples in scintillation vials were reweighed and lipid content determined by difference. Based on the lipid weight, samples were diluted in different volumes of isopropanol (range 2 to 8 ml). The diluted samples were then used for triglyceride analysis using Sigma kit 337-A (Sigma Chemical Co., St. Louis, MO) with isopropanol as a blank.

Liver Incubations

Upon return to the laboratory (usually within 1 h of biopsy), liver was sliced using a Krumdieck Tissue Slicer (Alabama Research and Development, Munford, AL) filled with ice-cold phosphate buffered (0.02 m*M*) saline (0.9% wt/vol). Slices of tissue (57 ± 11 mg) were then weighed into flasks. For measurement of conversion of [1-^{14}C]propionate (sodium salt; American Radiolabeled Chemicals, St. Louis, MO) to CO_2 and glucose, the incubation procedures of Overton et al. (1999) were used, except the Krebs-Ringer bicarbonate (KRB) media did not contain phenol red, and incubations were carried out for 120 min rather than 90 min. All chemicals were cell-culture tested or the highest purity available from Sigma Chemical Co., St. Louis, MO.

Conversion of [1-^{14}C]palmitate (sodium salt; American Radiolabeled Chemicals) to CO_2 was conducted according to the procedures described by Drackley et al. (1991b), except that the incubation medium was KRB (pH 7.4) with part of the NaCl (25 m*M*) replaced by sodium HEPES (6.5 g/L). To determine esterification of palmitate into intracellularly stored lipid compounds, the procedure of Drackley et al. (1991a) was used, with the exception of the incubation medium described above.

Determination of conversion of [1-^{14}C]propionate to glucose. After removing the septa during flask processing for CO_2 determination, 50 μl of an internal standard [1.1 μCi/ml [^{3}H]L-1-glucose (American Radiolabeled Chemicals) in bicarbonate-free KRB] were added to each flask to correct for recovery of glucose during processing. Flasks were swirled to mix the contents, and contents were poured into 50-ml centrifuge tubes containing 50 μl of Fisher Universal Indicator (Fisher Scientific, Pittsburgh, PA). Contents were made basic (pH ~ 9) through the addition of a saturated solution of $Ba(OH)_2$ in H_2O drop-wise. Tubes were then centrifuged at 915 × *g* for 10 min. This caused the formation of a pellet that allowed the supernatant to be poured into scintillation vials. Vials were capped and stored at –20°C until subsequent isolation of glucose.

Glucose (radiolabeled and nonradiolabeled) from metabolic incubations was isolated using the batch procedure of Azain et al. (1999), modified to allow for correction by the [^{3}H]L-1-glucose internal standard. Briefly, scintillation vial contents were thawed and poured into disposable 50-ml centrifuge tubes. Two milliliters of a 50% (wt/vol) slurry of an anion exchange resin (AG 1-X8 resin, acetate form; Bio-Rad, Hercules, CA) in water were added to each tube. Tubes were capped and vortexed vigorously for 30 s. This was repeated twice, allowing resin to settle momentarily between mixing bouts. Afterwards, 1.8 ml of a cation exchange resin (AG 50W-8 resin, hydrogen form: Bio-Rad, Hercules, CA) were added. The tubes were capped again and vortexed as above. Tubes were then centrifuged at 1390 × *g* for 15 min. The resins formed a solid pellet that allowed the aqueous layer to be poured into scintillation vials. The vials then were placed into a 37°C water bath and a steady stream of air was used to evaporate the water. The next day, 1 ml of water was added to reconstitute the glucose. Ten milliliters of scintillation cocktail (Scintisafe Econo 2, Fisher Scientific, Pittsburgh, PA) were added and vials were counted using dual label liquid scintillation spectroscopy (Model 2200 CA, Packard Instrument Company, Downers Grove, IL). An equal volume of the internal standard ([^{3}H]L-1-glucose) added to each flask was counted separately; therefore, correction of recovery based upon the internal standard was calculated for each incubation flask.

Statistical Analysis

Before statistical analysis, daily measurements (DMI, milk production) were reduced to weekly means. For in vitro incubations, triplicate values for blanks and live flasks were averaged and blanks subtracted from values obtained from live flasks. Rates of substrate conversion were calculated based upon the known amount of radioactivity added to each flask that was converted to end products over the 2-h incubation period.

Data for two cows were eliminated before statistical analysis. One cow fed 0 g/d of RPC developed hydropsy and did not have a normal parturition. Another cow fed 0 g/d of RPC had severe *Staphylococcus aureus* mastitis and had very low milk yields. Therefore, only 10 cows were included in the statistical analysis for the 0 g/d RPC treatment. Two cows fed 45 g/d of RPC developed *Staphylococcus aureus* mastitis. Data were collected and utilized until these animals were removed from the study at 3 wk postpartum. Two cows fed 75 g/d of RPC developed illnesses unrelated to treatment, but the data for these cows up until the point of illness (d 23 and 47 postpartum) were included in the statistical analyses. The remaining data for these cows were treated as missing values.

Data were analyzed using the MIXED procedure of SAS (SAS Users Guide, 2001). Pretreatment BCS, previous lactation ME305, and pretreatment values for all measurements except milk yield and composition were used as covariables during the analysis. These covariables were removed from the model in a backward, stepwise fashion when not significant ($P > 0.15$). Week or day was used in the repeated statement with cow within treatment as the error term and the autoregressive covariate structure (Littell et al., 1998). Degrees of freedom were estimated using the Satterthwaite specification. Single degree of freedom contrasts were used to test the linear and quadratic effects of choline supplementation using adjusted coefficients due to the uneven spacing of treatments. Another contrast was used to determine the effect of choline supplementation versus the control. Least squares means are reported with the highest standard error for each variable. Significance was declared at $P \leq 0.05$ and trends at $0.05 < P \leq 0.15$. Only significant treatment by time interactions will be discussed.

RESULTS AND DISCUSSION

Diet compositions are presented in Table 2. Diets were similar to formulations except that the prepartum diet averaged 1.51 Mcal/kg of DM (NE_L), and the postpartum diet averaged 1.64 Mcal/kg of DM (NE_L). This was lower than the 1.64 and 1.68 Mcal/kg of DM (NE_L) that the diets were originally formulated to contain due to the variability in energy content of forages used over the course of this study. Methionine supply, expressed as a percentage of metabolizable protein as predicted by Cornell Penn Miner Dairy (version 1.0) was 2.06 and 2.14% for the prepartum and postpartum diets, respectively.

The incidence of health disorders is summarized in Table 4. Although the number of cows assigned to each treatment in this experiment are too few to permit appropriate statistical analysis, these data are reported due to the influence they may have had on subsequent data reported herein. A larger number of cows fed 60 g/d of RPC treatment seemed to have more difficult transitions, having numerically more incidence of displaced abomasum compared to the cows fed the other treatments. Based collectively upon the metabolic data reported throughout the remainder of this manuscript, this occur-

Table 4. Incidence of health disorders for cows fed four amounts of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum.

	RPC, g/d			
Item	0	45	60	75
Retained placenta	0	0	0	0
Displaced abomasum	0	1	3	1
Mastitis	2	0	1	3
Milk fever/hypocalcemia	2	0	2	1
Ketosis	1	0	2	1
Foot and leg problems	1	1	0	1
Twins	2	2	1	1

rence may have been random rather than attributable to treatment.

The DMI of cows fed the four dietary treatments was similar (Table 5) and averaged 12.4 kg/d during the 21-d prepartum period and 18.4 kg/d during the 63-d postpartum period. Because DMI did not differ among treatments, differences in production and metabolism in this experiment can be attributed to the presence or absence of RPC in the diet of cows. There was a quadratic tendency for milk yield such that cows fed 45 g/d of RPC had the greatest milk yield; cows fed 60 and 75 g/d of RPC were similar to cows fed 0 g/d of RPC ($P = 0.15$; Table 6). Although differences in percentage of milk fat among treatments were not significant, feeding RPC increased milk fat yield ($P = 0.05$) and tended to increase FCM yield ($P = 0.06$). Similar to milk yield, FCM yield tended to respond in a quadratic fashion. Because choline (as phosphatidylcholine) plays a role in fatty acid transport in blood, it is speculated that more lipoprotein triglycerides may have been available to the mammary gland for incorporation into milk fat. Neither percentage of milk CP nor yield was affected by treatment. There was a tendency for the concentration of TS to increase linearly as cows consumed greater amounts of RPC. The trend for increased milk fat yield and nonsignificant increases in milk CP yield led to a trend for increased yield of TS by cows fed RPC (Table 6). Urea N concentrations in milk decreased linearly as cows consumed increasing amounts of RPC (Table 6). This response can be attributed to changes in milk yield rather than a direct effect of RPC.

Pinotti et al. (2000) fed 20 g/d of RPC to cows starting approximately 14 d before calving through 30 d postpartum and found that cows consuming the RPC produced 3.5 kg/d more milk than the controls. There was no difference in milk yield during the 120-d period that RPC was fed to cows starting approximately 28 d before calving (Hartwell et al., 2000). They reported, however, that cows fed a low RUP (4% DM) diet supplemented with 12 g/d of RPC produced more milk. Milk yield was decreased when cows fed a high RUP (6.2% DM) diet were supplemented with 12 g/d of RPC. In previous research using cows in established lactation, postruminal infusion of choline chloride either increased (two experiments; 3.2 and 1.6 kg/d) or did not affect (one experiment) milk yield of dairy cows compared with the control infusion (Sharma and Erdman, 1989). Erdman and Sharma (1991) reported that cows fed 0, 0.078, 0.156, and 0.234% of the diet DM as RPC tended to produce more milk and more FCM as the amount of RPC fed was increased (maximum increase of 2.4 kg/d). Unlike our study, Hartwell et al. (2000) measured a decrease in milk protein content when cows were fed the low RUP diet in conjunction with increasing amounts of RPC.

Milk fatty acid composition was measured in order to examine more closely the effects of feeding RPC on milk fat. When fatty acid composition of milk is expressed as a proportion of total fatty acids (g/100g), only small tendencies for changes in some of the fatty acid concentrations are detectable (Table 7). Heptadecanoic acid (C17:0) declined in milk fat as the amount of RPC consumed by cows increased. A quadratic pattern was observed for C18:1, *trans* 11 such that cows fed 45 g/d RPC had milk fat with the lowest concentration of this fatty acid. There was a tendency for an interaction of treatment and week ($P = 0.13$) for the content of C18:1, *trans* 10 in milk fat such that milk fat from cows consuming

Table 5. Dry matter intakes of cows fed four amounts of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum.

	RPC, g/d					Contrasts[1]		
Item	0	45	60	75	SE	L	Q	Ch
Prepartum (−21 d to calving), kg/d	12.6	11.9	12.8	12.5	0.5	0.99	0.42	0.74
Postpartum (calving to 63 d in milk), kg/d	17.8	18.7	18.3	18.8	0.6	0.30	0.81	0.28

[1]Single degree of freedom contrasts were used to test linear (L) and quadratic (Q) effects of feeding cows increasing amounts of RPC. Contrasts were adjusted to reflect uneven spacing of treatments. A single degree of freedom contrast (Ch) was also included to test effects of feeding 45, 60, or 75 g/d of RPC compared to the control.

Table 6. Yield and composition of milk from cows fed four amounts of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum.[1]

	RPC, g/d					Contrasts[2]		
Item	0	45	60	75	SE	L	Q	Ch
Milk, kg/d	40.0	43.3	39.9	41.0	1.1	0.63	0.15	0.28
Fat, %	4.02	4.21	4.00	4.29	0.10	0.19	0.70	0.23
Fat, kg/d	1.58	1.78	1.56	1.71	0.05	0.16	0.22	0.05
3.5% FCM[3], kg/d	42.8	47.6	42.6	45.4	1.1	0.23	0.12	0.06
CP, %	2.97	3.02	3.01	3.09	0.06	0.21	0.60	0.32
CP, kg/d	1.17	1.28	1.18	1.24	0.04	0.30	0.37	0.16
TS, %	12.43	12.67	12.42	12.83	0.13	0.10	0.54	0.16
TS, kg/d	4.92	5.42	4.90	5.18	0.13	0.28	0.15	0.09
MUN[4], mg/dl	15.7	14.9	15.6	13.9	0.5	0.03	0.32	0.07

[1]Single degree of freedom contrasts were used to test linear (L) and quadratic (Q) effects of feeding cows increasing amounts of RPC.

[2]Contrasts were adjusted to reflect uneven spacing of treatments. A single degree of freedom contrast (Ch) was also included to test effects of feeding 45, 60, or 75 g/d of RPC compared to the control.

[3]FCM = 0.4324*(milk yield) + 16.2162*(fat yield).

[4]MUN = Milk urea N.

45 g/d of RPC had a much higher proportion at wk 8 compared with milk from cows fed the other three RPC treatments.

The yields of individual fatty acids in milk followed a similar pattern of overall milk fat yield for cows during the study (Table 8). This led to trends ($P < 0.15$) for

Table 7. Fatty acids (g/100 g of milk fat) in milk samples obtained during wk 2, 5, and 8 postpartum from cows fed four amounts of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum.

	RPC, g/d					Contrasts[1]		
Fatty acid[2]	0	45	60	75	SE	L	Q	Ch
C4:0	5.99	6.39	5.83	5.91	0.39	0.83	0.46	0.90
C6:0	2.28	2.37	2.33	2.27	0.10	0.97	0.42	0.73
C8:0	1.09	1.13	1.14	1.10	0.07	0.79	0.58	0.67
C10:0	2.05	2.10	2.15	2.07	0.15	0.82	0.76	0.76
C12:0	2.11	2.16	2.26	2.15	0.15	0.70	0.75	0.65
C14:0	7.86	7.85	8.00	7.92	0.31	0.83	0.97	0.87
C14:1	0.51	0.55	0.59	0.55	0.03	0.19	0.53	0.17
C15:0	0.63	0.61	0.62	0.56	0.04	0.25	0.48	0.40
C16:0	26.14	25.62	26.00	25.95	0.57	0.82	0.63	0.68
C16:1	1.54	1.64	1.48	1.46	0.08	0.50	0.20	0.93
C17:0	0.53	0.51	0.49	0.49	0.02	0.04	0.84	0.07
C18:0	13.60	12.92	12.77	13.55	0.46	0.59	0.17	0.33
C18:1,t6–8	0.32	0.31	0.32	0.33	0.01	0.72	0.28	0.93
C18:1,t9	0.23	0.24	0.24	0.24	0.01	0.24	0.91	0.31
C18:1,t10	0.42	0.46	0.42	0.39	0.03	0.65	0.21	0.94
C18:1,t11	1.05	0.92	1.01	1.06	0.05	0.91	0.04	0.35
C18:1,c9	27.34	27.90	27.70	27.65	0.88	0.77	0.74	0.69
C18:2,c9,c12	3.09	2.97	3.12	3.05	0.11	0.91	0.62	0.73
C18:3,c9c12c15	0.34	0.33	0.35	0.35	0.01	0.47	0.41	0.75
CLA	0.36	0.34	0.37	0.35	0.02	0.82	0.52	0.60
De novo[3]	22.54	23.14	22.95	22.52	0.85	0.90	0.54	0.74
Preformed[4]	47.24	46.96	46.81	47.43	1.07	0.99	0.70	0.89
Total C16	27.66	27.30	27.47	27.39	0.57	0.73	0.82	0.68

[1]Single degree of freedom contrasts were used to test linear (L) and quadratic (Q) effects of feeding cows increasing amounts of RPC. Contrasts were adjusted to reflect uneven spacing of treatments. A single degree of freedom contrast (Ch) was also included to test effects of feeding 45, 60, or 75 g/d of RPC compared to the control.

[2]Fatty acids are designated by carbon (C) chain length, inclusion of a double bond, and location of cis (c), or trans (t) double bonds.

[3]De novo = fatty acids <16 carbons in length.

[4]Preformed = fatty acids >16 carbons in length.

Table 8. Yield (g/d) of fatty acids in milk samples obtained during wk 2, 5, and 8 postpartum from cows fed four amounts of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum.

	RPC, g/d					Contrasts[1]		
Fatty acid[2]	0	45	60	75	SE	L	Q	Ch
C4:0	94.98	120.12	92.55	105.90	9.13	0.57	0.27	0.29
C6:0	35.72	44.38	36.43	39.88	2.54	0.34	0.16	0.13
C8:0	16.93	21.06	17.63	19.18	1.43	0.33	0.25	0.16
C10:0	31.90	38.91	33.14	35.95	2.93	0.42	0.36	0.23
C12:0	32.83	40.06	34.61	37.30	2.81	0.31	0.33	0.17
C14:0	123.16	145.59	123.02	137.93	6.88	0.26	0.34	0.13
C14:1	7.98	10.13	8.93	9.69	0.65	0.07	0.29	0.04
C15:0	9.86	10.98	9.58	9.79	0.64	0.46	0.24	0.72
C16:0	413.66	477.74	404.98	455.87	21.39	0.35	0.44	0.19
C16:1	24.62	30.67	23.71	26.15	2.04	0.78	0.14	0.35
C17:0	8.45	9.40	7.81	8.56	0.47	0.81	0.36	0.80
C18:0	219.27	239.76	204.79	237.61	13.43	0.66	0.97	0.60
C18:1,t6–8	5.16	5.61	5.18	5.76	0.28	0.25	0.86	0.28
C18:1,t9	3.70	4.30	3.81	4.21	0.19	0.11	0.48	0.07
C18:1,t10	6.65	8.42	6.54	6.86	0.62	0.89	0.09	0.39
C18:1,t11	16.87	16.88	15.95	18.37	1.01	0.58	0.27	0.87
C18:1,c9	437.58	517.81	445.31	492.01	29.72	0.29	0.42	0.17
C18:2,c9,c12	49.34	54.20	49.89	52.93	2.49	0.41	0.60	0.30
C18:3,c9c12c15	5.46	6.09	5.61	6.10	0.26	0.14	0.75	0.12
CLA	5.74	6.11	5.79	6.11	0.30	0.48	0.87	0.45
De novo[3]	353.69	431.25	356.09	395.25	22.15	0.30	0.19	0.12
Preformed[4]	755.43	875.44	747.51	835.23	42.97	0.36	0.46	0.22
Total C16	438.08	509.23	428.12	481.64	22.86	0.37	0.37	0.19

[1]Single degree of freedom contrasts were used to test linear (L) and quadratic (Q) effects of feeding cows increasing amounts of RPC. Contrasts were adjusted to reflect uneven spacing of treatments. A single degree of freedom contrast (Ch) was also included to test effects of feeding 45, 60, or 75 g/d of RPC compared to the control.

[2]Fatty acids are designated by carbon (C) chain length, inclusion of a double bond, and location of cis (c), or trans (t) double bonds.

[3]De novo = fatty acids <16 carbons in length.

[4]Preformed = fatty acids >16 carbons in length.

increased yields when choline was fed to cows for some fatty acids (C6:0, C14:0, C18:1, trans 9, and C18:3) and significant ($P \leq 0.05$) increases for C14:1. A quadratic tendency was observed for both C16:1 and C18:1, *trans* 10 such that cows consuming 45 g/d RPC had higher concentrations of these fatty acids as compared to the other treatment groups. There also were tendencies for linear increases in C14:1, C18:1, *trans* 9, and C18:3 as cows consumed more RPC.

In general, short- and medium-chain fatty acids (C4:0 – C14 and half of the C16) are synthesized in the mammary gland (Mansbridge and Blake, 1997); the remainder of the C16 fatty acids and the long-chain fatty acids are extracted from the bloodstream as preformed fatty acids originating from dietary sources, adipose tissue, or VLDL. During early lactation, de novo synthesis of fatty acids is inhibited by the high uptake of long-chain fatty acids (Bauman and Davis, 1974). As reviewed by Palmquist et al. (1993), the lower availability of malonyl-CoA for addition of acetyl units is primarily responsible for this decrease in de novo synthesis. Looking at the groupings of those fatty acids synthesized by the mammary

Table 9. Body weights and condition scores from cows fed four amounts of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum.

	RPC, g/d					Contrasts[1]		
Item	0	45	60	75	SE	L	Q	Ch
BW, kg	623	631	629	626	7.0	0.65	0.44	0.49
BCS[2]	2.93	3.01	3.02	3.01	0.07	0.31	0.65	0.28

[1]Single degree of freedom contrasts were used to test linear (L) and quadratic (Q) effects of feeding cows increasing amounts of RPC. Contrasts were adjusted to reflect uneven spacing of treatments. A single degree of freedom contrast (Ch) was also included to test effects of feeding 45, 60, or 75 g/d of RPC compared to the control.

[2]BCS: 1 = thin and 5 = fat (Wildman et al., 1982).

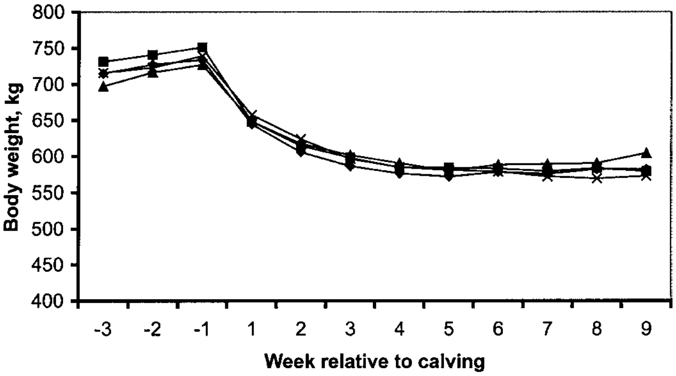

Figure 1. Body weights of cows fed either 0 g/d (◆), 45 g/d (■), 60 g/d (▲), or 75 g/d (✖) of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum. Weights obtained before assignment to treatment at approximately 28 d before the expected calving date of each cow were used during analysis of covariance. Effects of treatment (linear, $P = 0.65$; quadratic, $P = 0.44$; choline, $P = 0.49$) and treatment by time ($P = 0.72$) were not significant (SEM = 10.8).

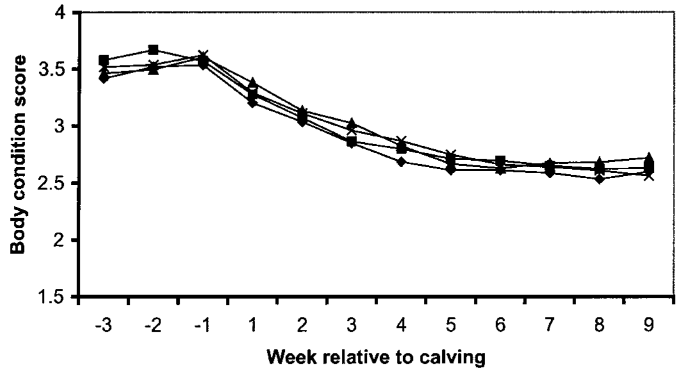

Figure 2. Body condition scores of cows fed either 0 g/d (◆), 45 g/d (■), 60 g/d (▲), or 75 g/d (✖) of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum. Body condition scores obtained before assignment to treatment at approximately 28 d before the expected calving date of each cow were used during analysis of covariance. Effects of treatment (linear, $P = 0.31$; quadratic, $P = 0.65$; choline, $P = 0.28$) and the interaction of treatment and time ($P = 0.74$) were not significant (SEM = 0.10).

gland and those taken up by the mammary gland and incorporated into milk fat show that RPC tended to increase de novo synthesis (choline effect; $P = 0.12$). It should be noted that cows fed RPC also had numerically higher yields of preformed fatty acids ($P = 0.22$). Because yields of all fatty acids were modestly affected by feeding RPC to cows, it is possible that availability of fatty acids from the bloodstream through VLDL is increased, and perhaps choline influences a particular mechanism in the mammary gland, such as milk fat globule formation. This clearly is speculative, but it provides an explanation for milk fat responses to choline reported in lactating cows in this study and others (Erdman and Sharma, 1991).

Body weights and BCS were not affected by treatment (Table 9, Figures 1 and 2). Hartwell et al. (2000) demonstrated that cows consuming 12 g/d of RPC had greater weight loss through 28 DIM than cows consuming either 0 or 6 g/d of RPC. Cows fed 0, 0.078, 0.156, or 0.234% RPC (0, 16.9, 36.2, and 51.2 g/d RPC; 0, 4.2, 9.1, 12.8 g/d choline chloride) on a DM basis starting 5 wk postpartum did not differ in BW (Erdman and Sharma, 1991).

Plasma concentrations of NEFA were similar among cows fed the four treatments (Table 8). Given that DMI was similar across treatments, this would indicate that choline does not affect release of fatty acids from adipose tissue during the periparturient period. It is also possible that any changes in NEFA release from adipose tissue are counterbalanced by increased uptake by the mammary gland. Hartwell et al. (2000) reported that feeding RPC did not affect plasma concentrations of NEFA. Other researchers (Pinotti et al., 2000) reported that cows fed 20 g/d of RPC had decreased circulating concentrations of NEFA on the day of parturition compared to controls (Pinotti et al. 2000). In both these studies, blood sampling was far too infrequent (only six and four times during the studies, respectively) to realistically assess effects of choline supplementation on plasma metabolites. Differences in plasma concentrations of βHBA among treatments in our experiment were not significant (Table 10), suggesting that choline supplementation did not affect hepatic ketogenesis directly. There was a tendency for a treatment × time interaction for plasma concentrations of βHBA. This was a result of cows fed 60 g/d RPC having higher concentrations of βHBA at 28 d

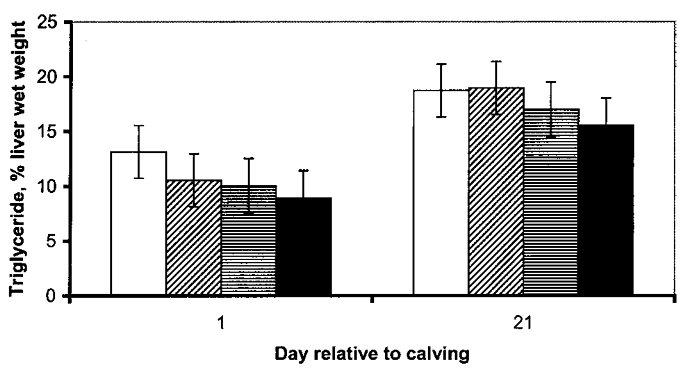

Figure 3. Concentrations of triglyceride in liver from cows fed either 0 g/d (white bars), 45 g/d (diagonal patterned bars), 60 g/d (horizontal patterned bars), or 75 g/d (black bars) of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum. Values from liver samples obtained before assignment to treatment at approximately 21 d before the expected calving date of each cow were used during analysis of covariance (4.73 +/– 0.41). Effects of treatment (linear, $P = 0.18$; quadratic, $P = 0.66$; choline, $P = 0.28$) and the interaction of treatment and time ($P = 0.91$) were not significant.

Table 10. Concentrations of metabolites in plasma obtained from cows fed four amounts of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum.

	RPC, g/d					Contrasts[1]		
Item	0	45	60	75	SE	L	Q	Ch
NEFA, μeq/L	367	425	328	365	27	0.67	0.18	0.84
Prepartum	130	157	116	145	21	0.81	0.73	0.67
Postpartum	579	640	514	547	41	0.40	0.25	0.81
BHBA, mg/dl	11.9	12.5	13.4	13.9	1.7	0.35	0.76	0.46
Prepartum	6.6	7.2	6.6	7.7	0.4	0.16	0.73	0.21
Postpartum	16.5	17.0	19.0	19.3	2.5	0.37	0.77	0.49

[1]Single degree of freedom contrasts were used to test linear (L) and quadratic (Q) effects of feeding cows increasing amounts of RPC. Contrasts were adjusted to reflect uneven spacing of treatments. A single degree of freedom contrast (Ch) was also included to test effects of feeding 45, 60, or 75 g/d of RPC compared to the control.

postpartum and is probably associated with the numerically higher incidences of displaced abomasums and ketosis (Table 4) for this treatment.

Concentrations of liver triglycerides (Figure 3) on the day after calving were similar to other studies (Grum et al., 1996, Gruffat et al., 1997). By wk 3 or 4, concentrations of triglyceride have been shown to decrease in these studies, but the cows in the current study continued to accumulate triglyceride in the liver. The net energy of the diet for the one study (1.76 Mcal/kg; Grum et al., 1996) was considerably higher than that fed in this study (1.64 Mcal/kg). The intake of energy may contribute to differences in concentrations of liver triglycerides at 21 d postpartum because DMI was similar between cows on the two studies. Concentrations of liver triglyceride were similar between treatment groups ($P > 0.15$). Likewise, there was no effect of RPC on concentrations of liver triglyceride in the study of Hartwell et al. (2000). These authors did report, however, that initial BCS at 28 d prepartum was related to the extent that triglycerides accumulated in the liver postpartum. Because treatment assignments were random except for stratification by calving date, initial BCS was not removed as a confounding factor in their experiment. This may have lead to the variations in BW loss observed in their study as well.

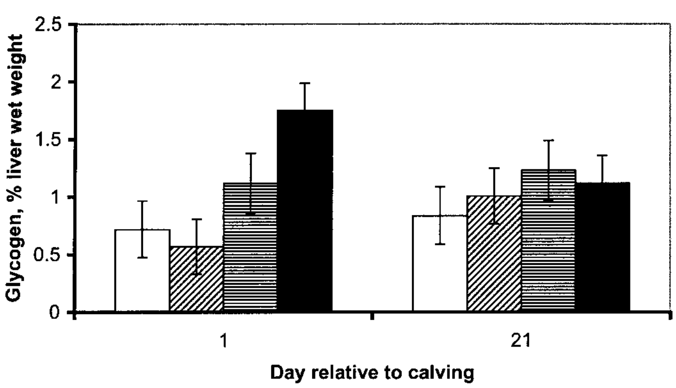

Figure 4. Concentration of glycogen in liver from cows fed either 0 g/d (white bars), 45 g/d (diagonal patterned bars), 60 g/d (horizontal patterned bars), or 75 g/d (black bars) of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum. Values from liver samples were obtained before assignment to treatment at approximately 21 d before the expected calving date of each cow were used during analysis of covariance (4.10 +/– 0.90). Glycogen content increased linearly ($P = 0.02$) in liver as cows consumed increasing amounts of RPC. There also was a tendency for a treatment by time effect ($P = 0.10$).

Liver concentrations of glycogen increased linearly as cows consumed increasing amounts of RPC ($P = 0.02$; Figure 4). Because Cardoniga-Valino et al. (1997) suggested that fatty liver inhibits gluconeogenesis, the slightly lower triglyceride concentrations for cows that received increasing amounts of RPC may have allowed for greater rates of gluconeogenesis, sparing glycogen from hydrolysis for use as a glucose source or perhaps replenishing glycogen in liver more quickly than cows not receiving RPC.

The capacity of liver slices to convert [1-^{14}C] palmitate to CO_2 in vitro was not affected by the amount of choline consumed by cows (Figure 5). This would indicate that hepatic capacity to conduct the first round of mitochondrial and peroxisomal oxidation of NEFA is not sensitive to choline supply in the periparturient dairy cow. The rate of conversion of [1-^{14}C] palmitate to esterified products stored intracellularly within the liver slices tended to decrease linearly ($P = 0.06$) as cows were fed increasing amounts of RPC. The value for liver slices from cows fed 75 g/d of RPC was 82% of that from cows receiving 0 g/d of RPC. Because palmitate concentrations were equivalent among flasks and hepatic capacity for NEFA oxidation was not affected significantly by treatment, these data suggest that feeding RPC to cows in this experiment increased the rate of triglyceride export from the liver as VLDL during the periparturient period. If choline does limit VLDL synthesis and secretion in periparturient dairy cows, this would concur with the results found in studies using either rat hepatocytes (Yao and Vance, 1988) or hepatic cell lines (Vermeulen et al., 1997), when

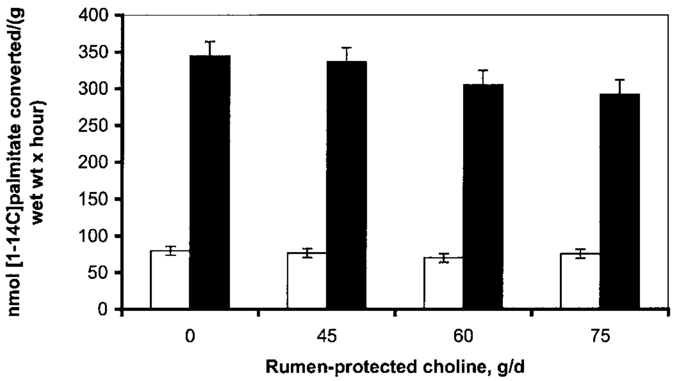

Figure 5. Conversion of [1-^{14}C]palmitate in liver slices in vitro to CO_2 (white bars) or stored esterified products (black bars). Liver was obtained from cows fed either 0 g/d, 45 g/d, 60 g/d, or 75 g/d of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum. Values from liver samples obtained before assignment to treatment at approximately 21 d before the expected calving date of each cow were used during analysis of covariance (74.8 +/– 19.2 and 175.1 +/– 32.8 for rates of [1-^{14}C]palmitate oxidation to CO_2 and cellular esterified products). The treatment (linear, P = 0.40, quadratic, P = 0.73; choline, P = 0.39) and treatment by time interaction (P = 0.22) for hepatic capacity to convert [1-^{14}C]palmitate to CO_2 were not significant. Hepatic capacity to convert [1-^{14}C]palmitate to stored esterified products tended to decreased linearly (P = 0.06) as cows were fed increasing amounts of RPC. The treatment by time interaction was not significant (P = 0.54).

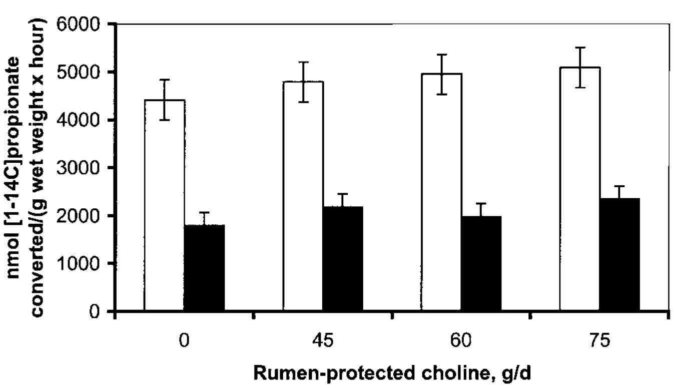

Figure 6. Conversion of [1-^{14}C]propionate in liver slices in vitro to CO_2 (white bars) or glucose (black bars). Liver samples were obtained from cows fed either 0 g/d, 45 g/d, 60 g/d, or 75 g/d of rumen-protected choline (RPC) from 21 d prepartum through 63 d postpartum. Values from liver samples obtained before assignment to treatment at approximately 21 d before the expected calving date of each cow were used during analysis of covariance. The mean covariable for [1-^{14}C]propionate conversion to CO_2 was 5007 +/– 1623 nmol/(g wet wt × h) and conversion to glucose was 2083 +/– 1104 nmol/(g wet wt × h). The treatment (linear, P = 0.23; quadratic, P = 0.96; choline, P = 0.27) and treatment by time interaction (P = 0.52) were not significant for the hepatic capacity to convert [1-^{14}C]propionate to CO_2. The treatment (linear, P = 0.21; quadratic, P = 0.94; and choline, P = 0.24) and treatment by time interaction (P = 0.25) were not significant for hepatic capacity to convert [1-^{14}C]propionate to glucose.

choline was supplemented in vitro in situations where the cells or donor animals were fed diets devoid of choline. Further research must be conducted to measure more directly hepatic capacity to export triglycerides as VLDL as affected by choline supply. Although detectable differences will be small, extracting lipids from the incubation media and separating the lipid classes by thin layer chromatography may be a useful tool for determining changes in VLDL secretion in vitro in future studies.

Hepatic capacities for conversion of [1-^{14}C] propionate to CO_2 or glucose as measured in this study were not different between treatment groups (P > 0.15; Figure 6). The linear increase in concentration of liver glycogen probably was not a result of increased gluconeogenesis unless the modest but nonsignificant increase in capacity for gluconeogenesis from propionate was sufficient to achieve this end.

Because effects of RPC are quite modest, feeding higher quantities during the periparturient period may have a greater impact on metabolic responses. In this study, concentrations of liver triglyceride and capacity of [1-^{14}C] palmitate conversion to stored esterified products were lowest, and concentrations of liver glycogen and rates of [1-^{14}C] propionate conversion to glucose were highest when cows consumed 75 g/d RPC. In contrast, cows that consumed 45 g/d RPC had the maximal response in milk and milk fat yield. Both hepatic and production responses may have been influenced by other factors not accounted for in this study and, therefore, a recommendation of an optimal dose of RPC cannot be made based on the results of this study.

CONCLUSIONS

Feeding cows RPC throughout the periparturient period resulted in decreases in the capacity of liver to accumulate NEFA as stored intracellular esterified products and increased glycogen content of liver. These metabolic changes likely explained the trends for increased yields of milk, milk fat, and 3.5% FCM of cows fed RPC during the periparturient period in this experiment.

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